Web3User Permission Lab

System Mindmap & Architectural Tree

Status snapshot: 3 November 2026
Development branch: `feature/m13-d-consensus`
Development HEAD: `25bfe62fbb1542e2f8512f522f9c8e4892b4ad97`

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1. Product vision and scope

- Explicit, deterministic, testable off-chain policy decisions for proposed Web3 actions.
- Decision model: ALLOW / DENY / UNKNOWN.
- Potential integrations: wallets, smart accounts, dApps, relayers, bundlers, and autonomous agents.
- No custody, EVM transaction signing, transaction broadcast, universal wallet, or production backend.

2. Authorization foundation — IMPLEMENTED

- ERC-20 `transfer(address,uint256)` evaluator and strict calldata validation.
- Canonical transaction and policy binding digests.
- Ed25519 V3 signed authorization artifacts with length-prefixed serialization.
- SQLite WAL persistence, single-use replay protection, policy/key governance.
- Cumulative budget enforcement.
- Monotonic clock, contention handling, safe TTL arithmetic, adversarial tests, process-crash recovery validation.

3. Witness / consensus evolution

- M11: monotonic witness adapter and fail-closed recovery — committed.
- M12 / M13-B: durable monotonic witness and bounded reconciliation — committed.
- M13-C / C.1: durable witness lifecycle integration and startup synchronization remediation — committed on main.
- M13-D: durable witness / consensus foundation — committed on development branch.
- Real-etcd integration harness: separate validation stage; pending report review and acceptance.

4. Verification and source protection

- Non-root Docker verification image — accepted.
- Isolated Compose PostgreSQL test harness — accepted.
- 655 / 655 Linux-compatible tests passed across 32 files.
- 26 Windows real-etcd tests excluded from the Linux run; native validation remains separate.
- Phase 1-A source-protection commit pushed: `25bfe62` .
- `.tmp/` binaries/fixtures and `.demo-data/` SQLite files were excluded from the commit.

5. Production readiness — BLOCKED

- Revocation freshness worker and synchronization validation.
- Canonical receipt preimage compatibility across gRPC and Engine.
- AWS KMS Witness Signer adapter and contract tests.
- Air-gapped backup-key ceremony.
- PostgreSQL / etcd / SQLite cross-store recovery and reconciliation.
- RPO / RTO restore drills.

- Independent security audit.

Architectural mindmap

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WEB3USER PERMISSION LAB
|
+-- Product boundary
|   +-- Off-chain policy decision SDK
|   +-- ALLOW / DENY / UNKNOWN
|   +-- No custody / no EVM signing / no broadcast
|
+-- Authorization foundation [IMPLEMENTED]
|   +-- ERC-20 transfer evaluator
|   +-- Canonical transaction binding
|   +-- Ed25519 V3 authorization artifacts
|   +-- SQLite WAL lifecycle and replay protection
|   +-- Policy/key governance and cumulative budgets
|   +-- Clock, contention, TTL, adversarial hardening
|
+-- Witness / consensus [IN PROGRESS]
|   +-- M11 monotonic witness adapter [COMMITTED]
|   +-- M12 / M13-B reconciliation pipeline [COMMITTED]
|   +-- M13-C / C.1 durable witness integration [MAIN]
|   +-- M13-D consensus foundation [DEV BRANCH]
|   +-- Real-etcd integration harness [PENDING ACCEPTANCE]
|
+-- Verification and source protection
|   +-- Non-root Docker harness [ACCEPTED]
|   +-- Isolated Compose PostgreSQL [ACCEPTED]
|   +-- 655 Linux-compatible tests [PASSED]
|   +-- 26 Windows real-etcd tests [SEPARATE VALIDATION]
|   +-- Phase 1-A source commit [PUSHED]
|
+-- Production readiness [BLOCKED]
    +-- Revocation freshness worker
    +-- Receipt preimage compatibility
    +-- AWS KMS signer adapter
    +-- Air-gapped backup key ceremony
    +-- Cross-store recovery reconciliation
    +-- RPO / RTO restore drills
    +-- Independent security audit
```

Status legend: Committed = present in Git history. Accepted = validation evidence reviewed. Production-ready = all security and operational gates closed.